Foundation Vision Models for Robotics: Understand, Compare, and Use

Foundation of Embodied AI · Concise course notes · October 18, 2026

> **Three questions only:** What does each family of foundation vision models actually output? How do SAM, FoundationPose, FoundationStereo, and SAM 3D differ? How should a roboticist select and combine these existing tools?
>
> **Scope:** Concepts, model selection, system composition, and limitations. Camera-calibration derivations, SE(3) filtering mathematics, policy training, code, and laboratory exercises are intentionally out of scope.

1. Start with the task: what visual output does the robot need?

A vision model is not an interchangeable perception black box. Select the **variable that the downstream robot needs** before comparing model size, benchmark ranks, or inference speed.

Task	Typical output	Question it answers	What it does not directly provide
Detection	Class label, bounding box, confidence score	"Where is the target region?"	Accurate contour or 3D distance
Instance segmentation / tracking	Pixel mask, object ID, temporal association	"Which pixels belong to this tool?"	Object depth or rigid 6D pose
Depth estimation	Depth map or stereo disparity	"How far is the visible surface?"	Full object geometry or an object frame
3D reconstruction	Estimated mesh, shape, or texture	"What might the object look like in 3D?"	Guaranteed metric dimensions or graspability
6D pose estimation	3D translation plus 3D rotation	"Where is the tool frame and how is it oriented?"	Contact force, friction, or grasp success

Four distinctions are essential: a mask is not depth; a depth map is not a complete 3D model; a shape model is not its current pose; and 6D pose is not contact state. The corresponding tools solve different problems.

Ask four questions first: Is locating the object enough, or is orientation required? Do you have RGB-D or a calibrated stereo pair? Do you have CAD or reference images? Is the result offline, or must it participate in real-time control?

2. Understand and distinguish the major models

2.1 Core model comparison

Model	Required inputs	Principal output	Best use and important boundary
SAM 2	Image/video, point/box/mask prompts	Instance masks propagated through video	**Track a specified object**; no 6D object pose.
SAM 3 / 3.1	Image/video, short text concepts, exemplars, or visual prompts	Open-vocabulary detection, masks, video tracks	**Find objects by language**; 3.1 improves multi-object video efficiency, not a universal real-time guarantee.
FoundationStereo	Calibrated, synchronized, rectified left/right images	Dense disparity convertible to metric depth	**Stereo geometry**; neither monocular depth nor object pose.
Fast-FoundationStereo	Same stereo prerequisites	Lower-latency stereo disparity	**Stereo under tighter latency budgets**; profile on the deployment GPU.
FoundationPose	Typically RGB-D, target mask, and CAD or reference images	Rigid-object 6D pose and tracking	**Initialize/update tool pose**; depends on geometry priors and good observations.
SAM 3D Objects	Object region in a single image	Candidate 3D geometry, appearance, and layout	**Shape prior without CAD**; a generated mesh is not certified engineering geometry.
SAM 3D Body	Human image, optional keypoints/masks	Human 3D body mesh and pose	**Human motion and shape**; not hand–object contact force.

2.2 Three distinctions that prevent most selection errors

SAM 2 versus SAM 3: "track a specified instance" versus "find instances matching a concept." If a click or bounding box already specifies the target, SAM 2 can propagate its mask through a video. If the instruction is "find every red screwdriver," SAM 3 can use a short textual concept or example. SAM 3.1 improves efficiency when processing multiple video objects. For long, compositional instructions, a VLM or agent may still need to reduce the request to a localizable concept.

FoundationStereo versus monocular depth: their geometric scale comes from different sources. FoundationStereo uses a **real, calibrated stereo pair** to estimate disparity and, with camera parameters, metric

depth. Monocular approaches (including selected Depth Anything 3 variants) infer geometry from images, but relative-depth checkpoints must not be confused with specialized metric-depth checkpoints. Neither should be assumed accurate enough for close-tolerance manipulation without validation. A head camera and a moving wrist camera are not automatically a valid rectified stereo pair.

FoundationPose versus SAM 3D Objects: "where is the rigid object now?" versus "what might its shape be?" FoundationPose uses a CAD model or reference imagery as an object prior to initialize and track pose from observations. SAM 3D Objects can produce plausible candidate geometry without existing CAD; its scale, coordinate frame, local shape, and suitability for precision tasks still need independent checks. It is not an automatic replacement for scanned, measured, or verified CAD.

2.3 Two useful complementary tools

- **Grounding DINO** is text-conditioned open-vocabulary object detection, primarily returning boxes. It is useful when boxes suffice or when chaining a separate segmentation model; its interface differs from the masks returned by SAM 3.
- **Depth Anything 3** supports several monocular and multi-view depth/geometry settings. The exact checkpoint matters: check whether the output is relative depth, metric depth, or multi-view geometry. It can provide a useful prior without a hardware depth sensor, but not an assumed substitute for calibrated, accurate RGB-D.

3. Choose and combine models for a robot task

3.1 A minimal selection procedure

Step 1 — Demand the minimum sufficient output. Stop at a detection or mask if the task only needs target presence and tracking. Add depth when distance matters, and rigid 6D pose only when orientation or geometric alignment is necessary. A direct vision-action policy (VLA) may generate actions directly from images and proprioception without explicitly estimating pose.

Step 2 — Check your sensors and object prior. If reliable aligned RGB-D is available, use it before adding another depth estimator. If you have calibrated synchronized stereo, consider FoundationStereo or Fast-FoundationStereo. With monocular images alone, treat predicted geometry and

scale cautiously. Use accurate CAD when available; otherwise consider multi-view reference images or SAM 3D Objects as a candidate shape source.

Step 3 — Build the smallest pipeline that meets the task. Not every foundation model needs to run sequentially on every frame. Language-based localization may run once at initialization and again on failure; lighter tracking and pose updates can handle intermediate frames.

3.2 Two representative robotics pipelines

Pipeline A — Explicit geometric state for tool grasping, orientation, and assembly.

> Language goal → **SAM 3** for localization and segmentation → **target mask + RGB-D + CAD/reference views** → **FoundationPose** for 6D pose initialization/tracking → robot controller; re-localize after tracking failure.

Choose this pipeline when the controller needs an object coordinate frame or geometric alignment errors. **SAM 3 does not itself deliver 6D pose; FoundationPose cannot conjure accurate CAD or missing depth.**

Pipeline B — Direct visual policies for VLA and closed-loop manipulation.

> Task language + camera images + robot proprioception → **VLA / visual policy** → robot actions; optional **SAM 3, tracker, or geometric estimator** for object monitoring, task verification, and recovery.

This pipeline need not include FoundationPose. The approaches can be hybridized: retain explicit geometry for supervision or safety checks while leaving the policy's primary action interface unchanged.

3.3 Four practical situations

Robot question	Minimal suitable choice	When to extend it
“Find the red screwdriver on the table.”	SAM 3 for a language-conditioned mask; keep tracking in video.	Add depth and pose estimation if grasp orientation matters.

Robot question	Minimal suitable choice	When to extend it
“Pick up a tool from arbitrary orientations and rotate it to a target pose.”	RGB-D + mask + CAD/reference imagery + FoundationPose.	Re-localize after occlusion/drop; use additional feedback for contact.
“I have no CAD model for this unfamiliar object.”	SAM 3D Objects for a candidate mesh.	Verify scale, surface detail, and shape before precision control.
“I have only monocular RGB but want scene distance.”	Appropriate Depth Anything 3 variant for a geometric prior.	Add scale constraints, RGB-D, or calibrated stereo for reliable metric control.

4. Practical usage: what must be checked?

4.1 Six pre-deployment checks

1. **Input contract:** Does this specific model require RGB, RGB-D, stereo, CAD, reference images, text, boxes, or masks? Do you actually have them?
2. **Scale and frames:** Is depth in meters? Are mask and depth registered? Is the reported pose in camera or robot-base coordinates? Camera extrinsics must be calibrated separately.
3. **Time and identity:** Is the same object ID maintained across frames? What triggers re-localization after occlusion? Is the timestamp from image capture or inference completion?
4. **Failure cases:** Which component will fail on transparency, reflections, weak texture, symmetry, high-speed motion, or a hand blocking the target?
5. **Latency budget:** End-to-end delay includes sensing, inference, communication, and control. Paper-level FPS is not the robot's control rate.
6. **Licenses and interfaces:** Check official code, weights, and commercial licenses; confirm data formats, output definitions, memory demands, and supported modes.

4.2 Tasks that vision foundation models do not solve alone

A confident mask does not establish an accurate pose; an accurate pose does not establish successful contact. Dexterous grasping, tight-tolerance insertion, and tool rotation may need joint proprioception, tactile/force feedback, and corrective control. When tracking becomes unreliable or an object falls, the system should be able to pause, re-observe, and re-initialize instead of acting on stale poses.

Evaluate the whole robot system, not only the model. A paper demonstrates model capabilities; actual robotic reliability additionally depends on sensor data, latency, calibration, object priors, and downstream control. A visually symmetric orientation that is unobservable in the image cannot be recovered merely by increasing model size.

4.3 Seven-line selection cheat sheet

- **Find an object from language → SAM 3;** need only open-vocabulary boxes → Grounding DINO.
- **Track an already identified object mask → SAM 2;** examine SAM 3.1 for multiple video objects.
- **Already have reliable RGB-D → use that depth;** do not add a model without a reason.
- **Calibrated synchronized stereo for depth → FoundationStereo;** tighter latency → Fast-FoundationStereo.
- **Monocular RGB for rough geometry → Depth Anything 3;** verify the checkpoint's scale semantics.
- **Have object CAD/reference views and need rigid 6D pose → FoundationPose.**
- **Need a candidate object shape without CAD → SAM 3D Objects;** use SAM 3D Body for human 3D reconstruction.

Official links and further reading

Capabilities and limitations above are based on the following official papers or repositories (all links are clickable):

SAM 2 · **SAM 3 / 3.1** · **SAM 3.1 release** · **FoundationPose** · **FoundationStereo** · **Fast-FoundationStereo** · **SAM 3D Objects** · **SAM 3D Body** · **Grounding DINO** · **Depth Anything 3**

Related courses: Camera intrinsics/extrinsics, transforms, and calibration → *Robot Camera Calibration*; demonstrations and action learning → *Data Collection and Imitation Learning*; visual closed-loop policies and control → *Embodied Reinforcement Learning* and *Advanced Robot Agents*. Those details are deliberately not repeated here.

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